

- Description

This module manages a smart road ice detection and anti-icing system, ensuring that:

1. **The road surface temperature is continuously monitored to detect ice formation risks.**
2. **If the temperature drops below freezing (0°C), anti-icing salt is automatically dispersed.**
3. **If the temperature rises above a safe threshold, salt dispersion is stopped to prevent waste.**
4. **A log of all temperature readings, salt dispersion events, and system activations is maintained for road safety optimization.**

This system helps reduce road accidents, improve driving safety, and optimize salt usage for cost efficiency.

- Specification

module SmartRoadIceDetection

var currentTemperature, freezingPoint, stopSaltDispersionTemp: real

var saltDispersionActive: boolean

process updateRoadTemperature(changeAmount: real)

functional scenario form:

currentTemperature >= freezingPoint && Road temperature maintained.

||

currentTemperature < freezingPoint && saltDispersionActive = true, Salt dispersion activated.

||

currentTemperature > stopSaltDispersionTemp && saltDispersionActive = false
Salt dispersion deactivated.

end process

end module

- Java Code

```
package code;
class SmartRoadSafety {
    private double freezingPoint;
    private double stopSaltDispersionTemp;
    private double currentTemperature;
    private boolean saltDispersionActive;
    public SmartRoadSafety(double freezingPoint, double stopSaltDispersionTemp, double initialTemperature) {
        this.freezingPoint = freezingPoint;
        this.stopSaltDispersionTemp = stopSaltDispersionTemp;
        this.currentTemperature = initialTemperature;
        this.saltDispersionActive = false;
    }
}
```

```

    public void updateRoadTemperature(double changeAmount) {
        currentTemperature += changeAmount;
        manageSaltDispersion();
        System.out.printf("Current Road Temperature: %.2f°C\n", current
Temperature);
        if (saltDispersionActive) {
            System.out.println("Anti-
Icing Salt Dispersion Activated.");
        } else {
            System.out.println("Anti-
Icing Salt Dispersion Deactivated.");
        }
    }
    private void manageSaltDispersion() {
        if (currentTemperature < freezingPoint) {
            saltDispersionActive = true;
        } else if (currentTemperature > stopSaltDispersionTemp) {
            saltDispersionActive = false;
        }
    }
}
public class c64_SmartRoadSafetySystem {
    public static void main(String[] args) {
        SmartRoadSafety roadSafety = new SmartRoadSafety(0.0, 5.0, 3.0)
;

        // testcase-VT:
        //     roadSafety.updateRoadTemperature(-4.0);
        //     roadSafety.updateRoadTemperature(3.0);
        //     roadSafety.updateRoadTemperature(4.0);
        //     roadSafety.updateRoadTemperature(-7.0);

        // testcase-FT:
        //     roadSafety.updateRoadTemperature(-3.1597835149418285E307);
        //     roadSafety.updateRoadTemperature(-1.7836486570745169E308);
        //     roadSafety.updateRoadTemperature(-4.9E-324);
        //     roadSafety.updateRoadTemperature(3.6603310227959246E298);

        // testcase-Z3:
        //     roadSafety.updateRoadTemperature(-4.0);
        //     roadSafety.updateRoadTemperature(3.0);
        //     roadSafety.updateRoadTemperature(4.0);
        //     roadSafety.updateRoadTemperature(-7.0);

        // testcase-UVT:

```

```

        roadSafety.updateRoadTemperature(-4.0);
        roadSafety.updateRoadTemperature(3.0);
        roadSafety.updateRoadTemperature(4.0);
        roadSafety.updateRoadTemperature(-7.0);
    }
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c64_SmartRoadSafetySystem.java	100.0 %	6	0	6
> c64_SmartRoadSafetySystem		0	0	0
> SmartRoadSafety	100.0 %	6	0	6

3. Vibration testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,684	20,770
▼ src	0.4 %	86	20,684	20,770
▼ code	0.4 %	86	20,684	20,770
▼ c64_SmartRoadSafetySystem.java	96.6 %	86	3	89
> c64_SmartRoadSafetySystem	87.0 %	20	3	23
> SmartRoadSafety	100.0 %	66	0	66

4. Fuzz testing branch coverage: 33.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c64_SmartRoadSafetySystem.java	33.3 %	2	4	6
> c64_SmartRoadSafetySystem		0	0	0
> SmartRoadSafety	33.3 %	2	4	6

5. Fuzz testing instruction coverage: 83.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	74	20,696	20,770
▼ src	0.4 %	74	20,696	20,770
▼ code	0.4 %	74	20,696	20,770
▼ c64_SmartRoadSafetySystem.java	83.1 %	74	15	89
> c64_SmartRoadSafetySystem	87.0 %	20	3	23
> SmartRoadSafety	81.8 %	54	12	66

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c64_SmartRoadSafetySystem.java	100.0 %	6	0	6
> c64_SmartRoadSafetySystem		0	0	0
> SmartRoadSafety	100.0 %	6	0	6

7. Z3 instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,684	20,770
▼ src	0.4 %	86	20,684	20,770
▼ code	0.4 %	86	20,684	20,770
▼ c64_SmartRoadSafetySystem.java	96.6 %	86	3	89
> c64_SmartRoadSafetySystem	87.0 %	20	3	23
> SmartRoadSafety	100.0 %	66	0	66

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c64_SmartRoadSafetySystem.java	100.0 %	6	0	6
> c64_SmartRoadSafetySystem		0	0	0
> SmartRoadSafety	100.0 %	6	0	6

9. Unimproved V-method instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,684	20,770
▼ src	0.4 %	86	20,684	20,770
▼ code	0.4 %	86	20,684	20,770
▼ c64_SmartRoadSafetySystem.java	96.6 %	86	3	89
> c64_SmartRoadSafetySystem	87.0 %	20	3	23
> SmartRoadSafety	100.0 %	66	0	66